

****Multiple Choice Questions:****

1. What is the definition of one mole of a substance?
 - a) The mass of a substance in grams.
 - b) The relative molecular mass of a substance.
 - c) 6.02×10^{23} particles of that substance.
 - d) The sum of the relative atomic masses of its constituent elements.

2. What term describes different arrangements of atoms for a given molecular formula?
 - a) Enantiomers
 - b) Isomers
 - c) Alkanes
 - d) Homologous series

3. What is a chiral center?
 - a) A carbon atom attached to three different groups.
 - b) A carbon atom attached to four different groups.
 - c) A carbon atom involved in a double bond.
 - d) A carbon atom involved in a triple bond.

4. What is the general molecular formula for alkenes?
 - a) C_nH_{2n+2}
 - b) C_nH_{2n}
 - c) C_nH_{2n-2}
 - d) C_nH_n

5. What type of hybridization do trigonal carbons (found in double bonds) exhibit?

- a) sp
- b) sp²
- c) sp³
- d) None of the above

****Free Response Questions:****

1. Explain the difference between structural isomerism and stereoisomerism, providing an example of each.
2. Describe the process of elemental analysis and how it is used to determine the percentage of each element in a compound.
3. Explain the concept of cracking, including the conditions under which it occurs and the products formed.

****Answer Key:****

****Multiple Choice:****

1. c

2. b

3. b

4. b

5. b